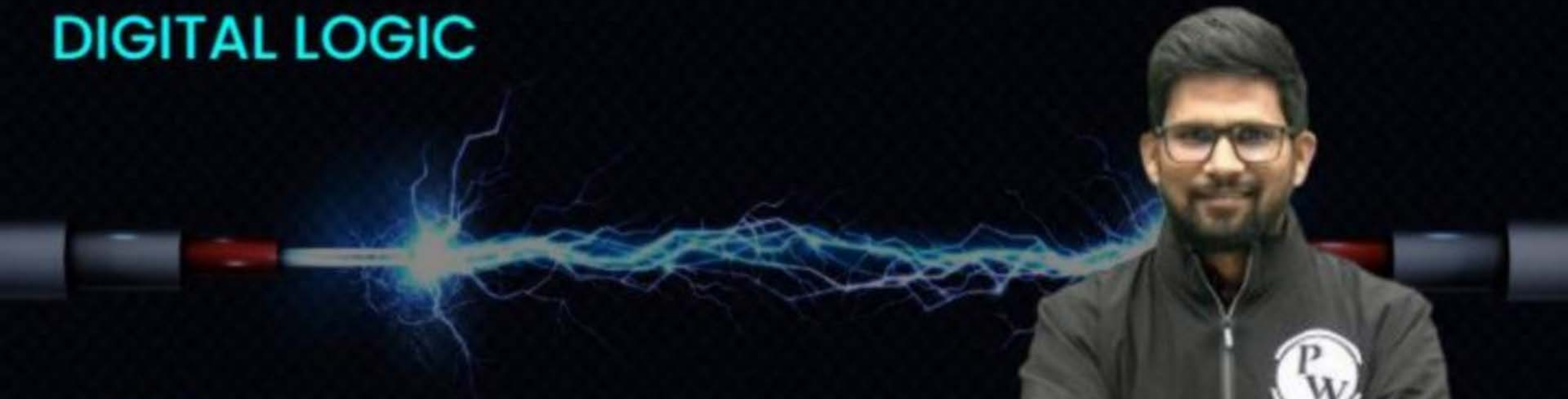
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No: 03

Sequential Circuit



By- Chandan Gupta Sir

Recap of Previous Lecture

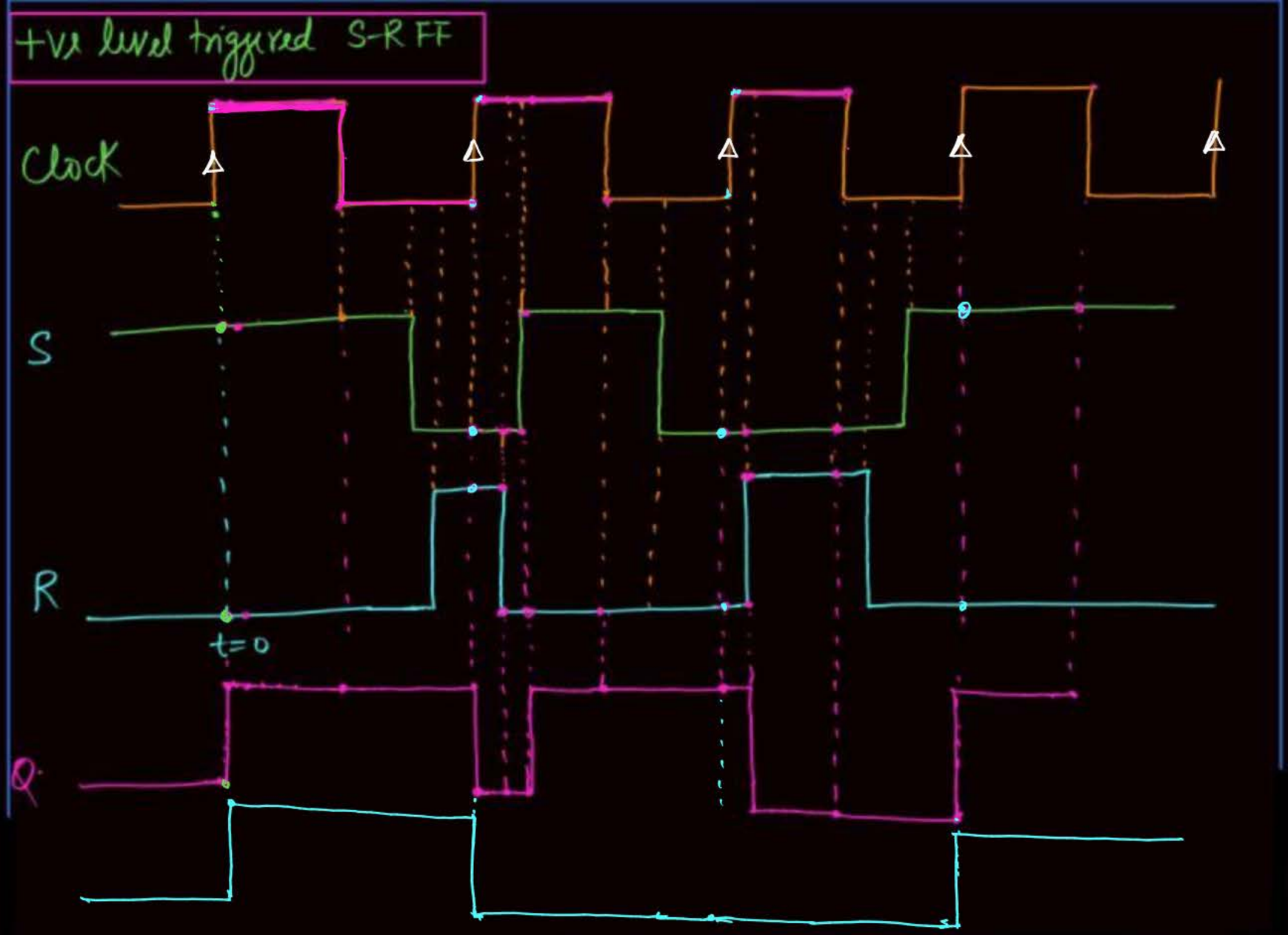


- J-K ff
- Level & Edge triggered
-



- Race Around condition
- Master - slave ff
- Counter

+ve level triggered S-R FF

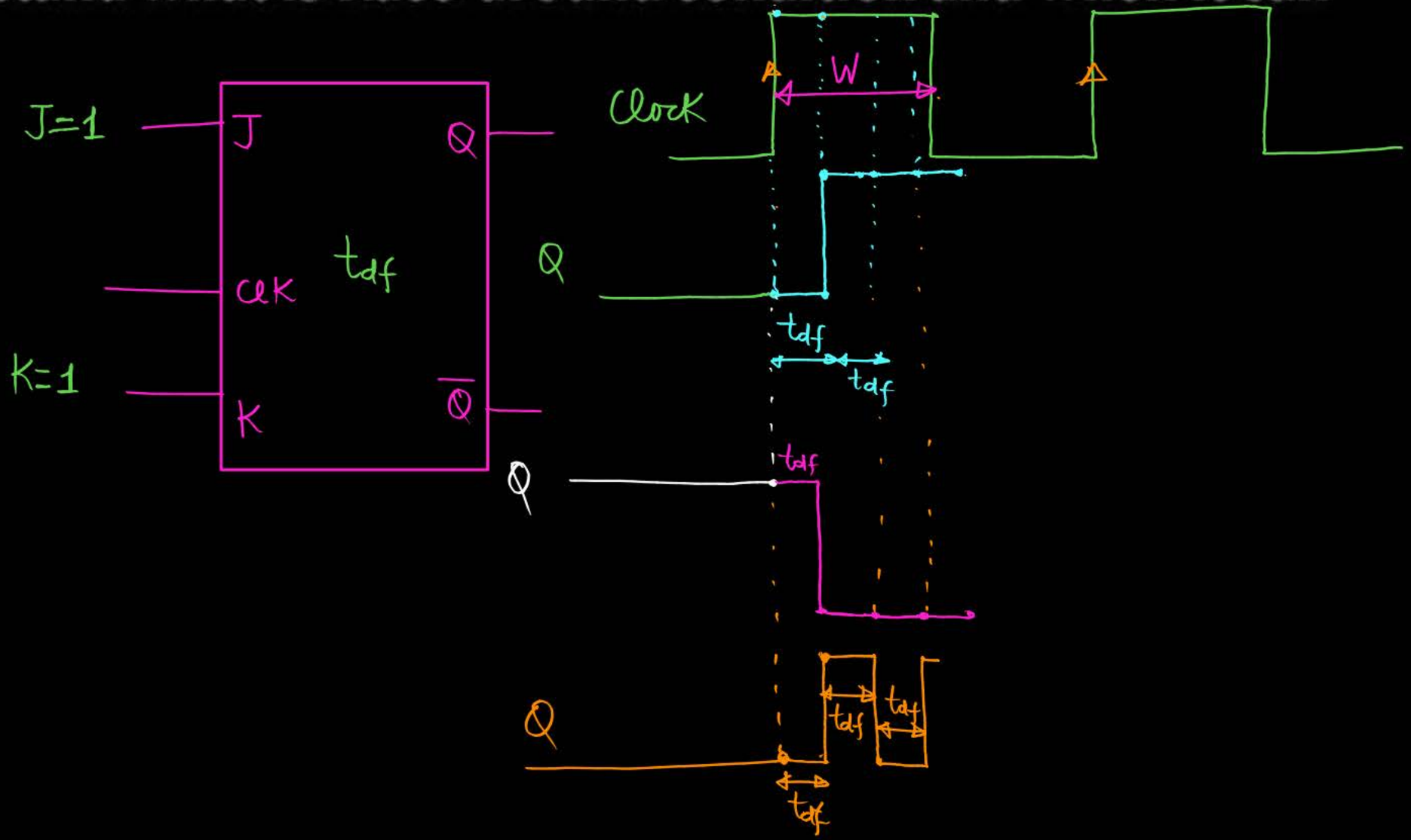


H.W

if ff is -ve level triggered then draw the O/p wave form

Race Around Condition

- Lets understand what is Race-around condition and when it can happen :



- In level triggered J-K ff with J=1 & K=1, if pulse width W >> t_{of} then O/P changes more than once in a single clock cycle. This condition is called as Race-Around condition.

- **Final condition for Race Around Condition :**

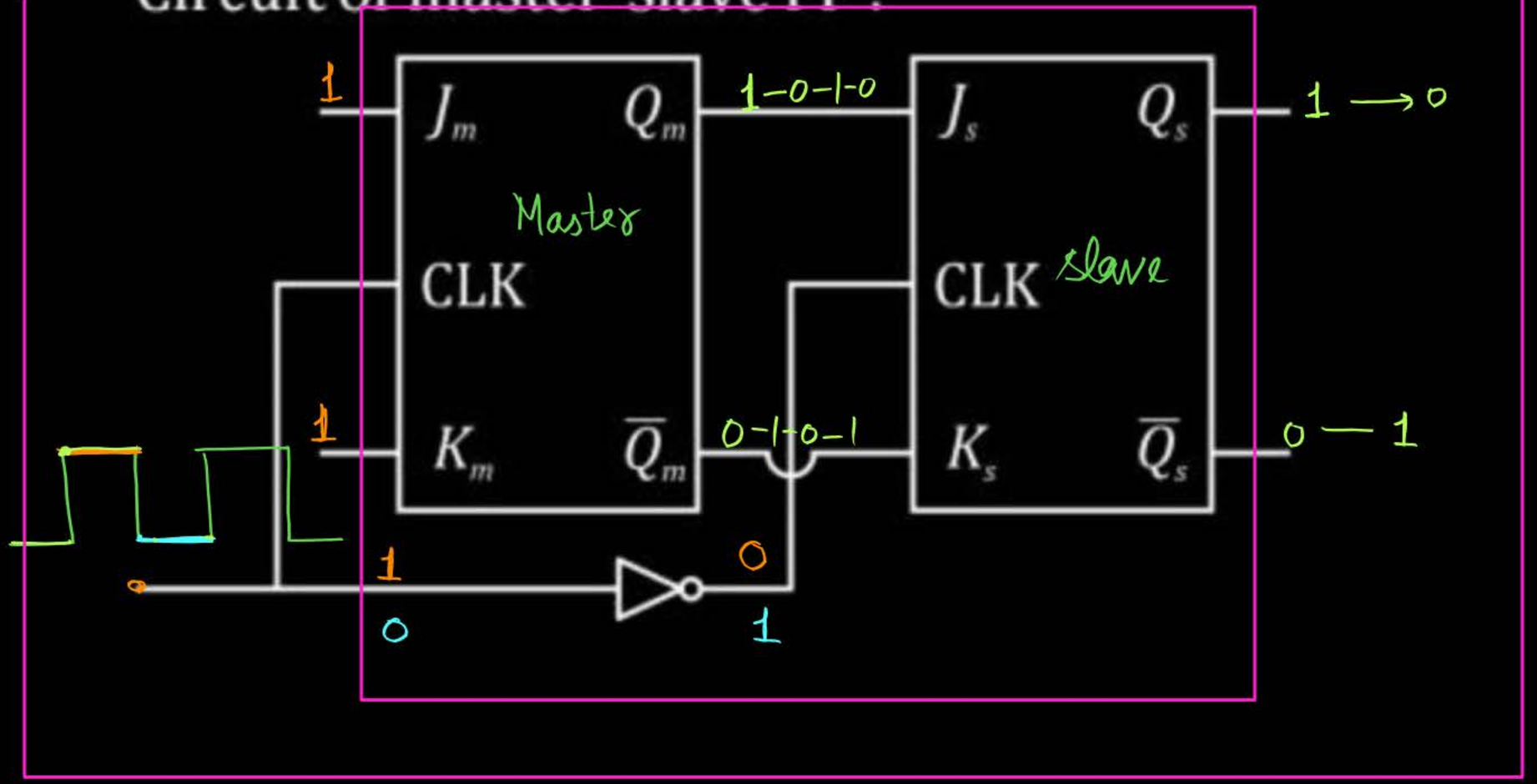
- FF must be J-K ff.
- I/P of J-K ff must be J=1, K=1.
- Level triggered J-K ff.
- Pulse width W >> t_{pf}.

→ If all of the above condition will satisfy then only Race-Around condition will happen.

Note : • Race-Around condition will never happen in edge-triggered J-K ff.
• Race-Around condition will never happen in S-R ff.

How to avoid race around condition using Master slave FF

- Circuit of master-slave FF :



How to avoid race around condition using Master slave FF

- Working :

- O/P of Master-slave ff is O/P of slave ff.
- In Master-slave ff, Master ff is level triggered and slave ff is edge triggered.
- Master-slave ff is used to avoid race-around condition.

Question

Which of the following is/are true ?

(a) Race around condition occurs when $J = K = 1$ and edge triggered is applied.

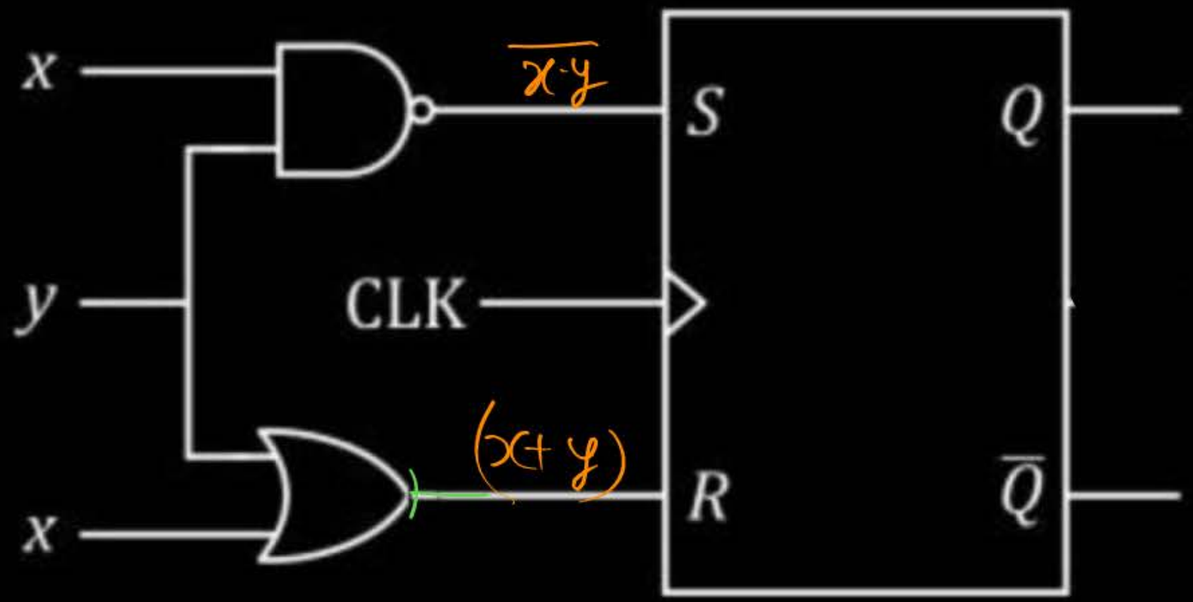
(b) Race around condition occurs when $J = 1, K = 0$ and level triggered is applied.

(c) Race around condition never occurs in edge triggered J-K FF.

(d) Race around condition occurs in S-R FF with $S = 0, R = 1$ and level triggered is applied.

[Question] MSQ

An S-R FF circuit is as given below :



Characteristic equation of above FF will be :

(a) $\bar{x}$

(b) $\bar{y}$

(c) $\overline{x \cdot y}$

(d) $\overline{x + y}$

$Q(n+1) = S + \bar{R}\,Q(n)$, $S \cdot R = 0$

$= \overline{x \cdot y} + \overline{x+y}\;Q(n)$

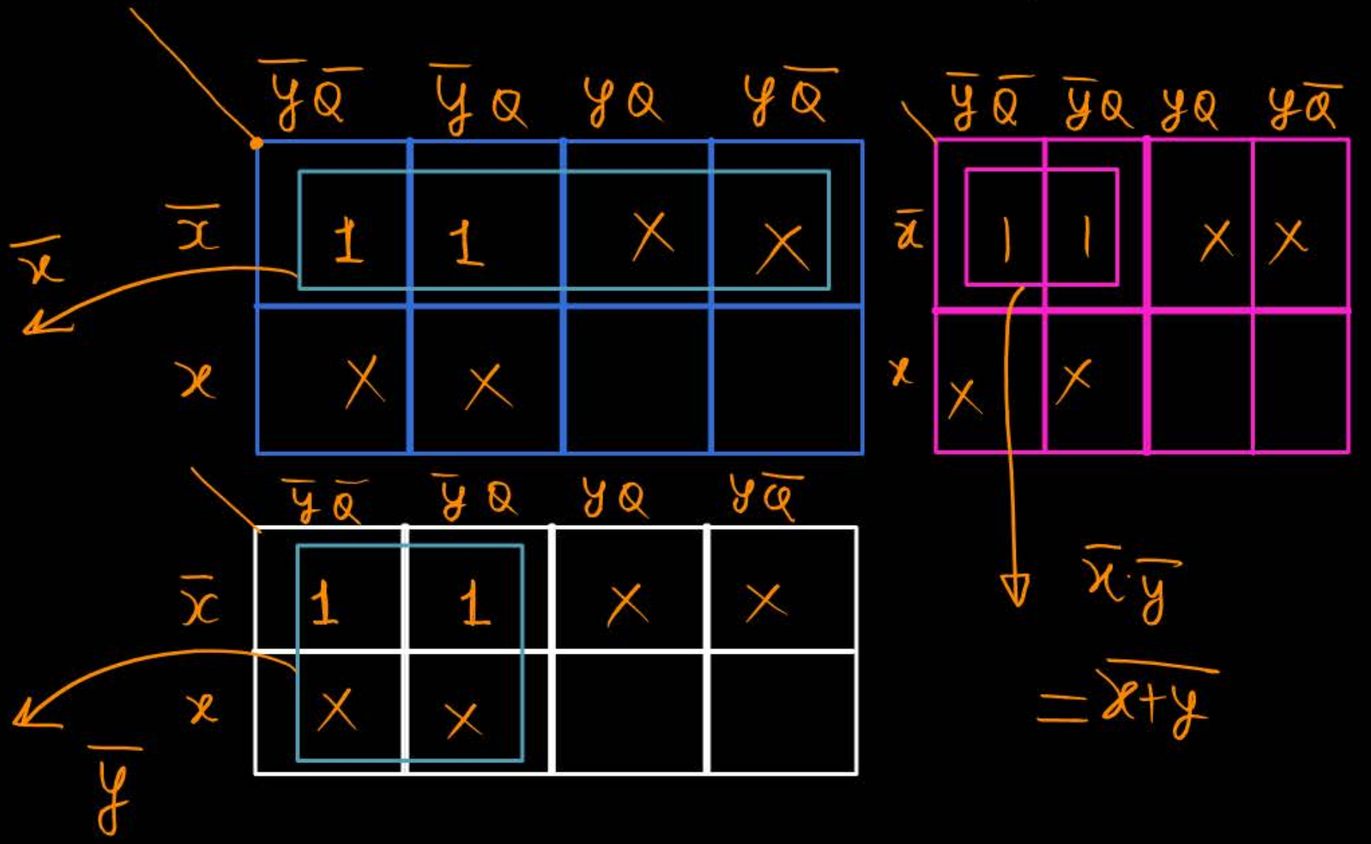
$= \bar{x} + \bar{y} + \bar{x}\bar{y}\,Q(n)$

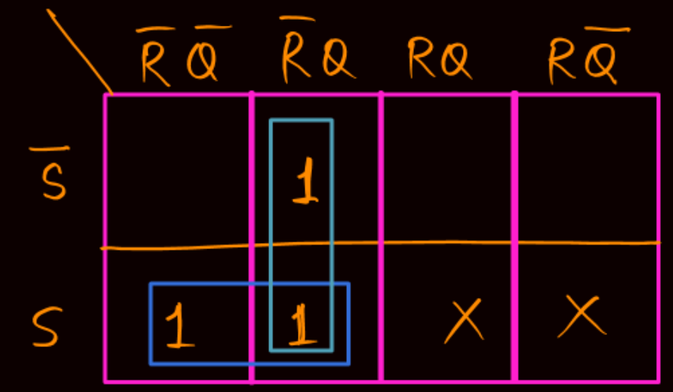
$= \bar{x} + \bar{y}$

$= \overline{x \cdot y}$

x	y	Q(n)	Q(n+1)
0	0	0	1
0	0	1	1
0	1	0	X
0	1	1	X
1	0	0	X
1	0	1	X
1	1	0	0
1	1	1	0

$$Q[n+1][x, y, Q(n)] = \Sigma(0,1) + d\Sigma(2,3,4,5)$$





$$Q(n+1) = S\bar{R} + \bar{R}Q(n)$$

Counters

- What is counter and for what it is used for ?
- To count clock pulses applied.

- MOD No. of the counter :

- The total no. of different states in a counter is called as MOD no of the counter.

- $0 - 1 - 3 - 2 - 4 - 7$ → MOD no = 6

- $1 - 2^1 - 2^2 - 2^3 - 2^4 \cdots\cdots 2^7$ → MOD no = 8

- $1 - 6 - 9 - 4 - 3 - 11 - 12 - 19 - 7 - 2$ → MOD no = 10

- $0 - \underline{1} - 2 - 1 - 3 - 1 - 4$ → MOD no = 7

- 0 - 1 - 2 - 3 - 4 - 1 - 2 - 3 - 5 - 1 - 2 - 3 → MOD no. = 12

- 1 - 2 - 2 - 3 - 3 - 3 - 4 - 4 - 4 - 4 → MOD no. = 10

Thank you

GW

Soldiers !

